

Grade 2 Ontario Curriculum Algebra

What's Missing?

Unit Overview: What's Missing?

Grade: Grade 2 | **Strand:** Algebra

Target Expectations: C2.1, C2.2, C2.3

Learning Goal: Children will identify symbols used as variables; determine what to add or subtract to make two expressions equivalent; and find equivalent relationships for whole numbers up to 100.

Lesson	Lesson Title	Learning Goal
Day 1	Symbols Stand In	I can recognise when a symbol stands for an unknown number.
Day 2	Missing Number in Addition	I can find the missing number in an addition equation.
Day 3	Missing Number in Subtraction	I can find the missing number in a subtraction equation.
Day 4	Many Ways to Make 50 (and 100)	I can find many equivalent expressions for the same whole number up to 100.
Day 5	Mystery Case Capstone	I can write a mystery case using a variable and solve it.

Lesson 1: Symbols Stand In

Goal: A box, a blank, or a letter can stand for an unknown number. We call it a VARIABLE.

Expectation: C2.1

Learning Goal: "I can recognise when a symbol stands for an unknown number."

Materials: M1_magnifying_cards, M9_variable_chart, M10_detective_poster

1. Minds On (12 mins): Detective Dee's First Case

Detective Dee found a clue: $5 + ? = 12$. What does the question mark stand for? Detective Dee opens: 'Variable, variable, hold the place — the unknown number fills the space!' (Class echoes.)

Bring children to the carpet. Detective Dee + Mystery Mo puppets. 'Friends — Detective Dee just got a new case. Let me read it.'

2. Action (25 mins): Variable-Hunt Stations

- Whole group: read 4 equations together.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: use a letter as a variable (n, x, or first initial).
- Discuss the big idea.
- Circulate to record formative evidence on the variable tracker.
- INVENT-AND-COMPARE: Two pairs show their Station C equations.

3. Consolidation (13 mins): Detective Dee's Variable Brag

Discuss:

- What does VARIABLE mean?
- Why do we use a variable instead of a question mark for everything?
- Can a letter be a variable?
- What's the difference between the variable and the answer?

Exit: Detective Dee's refrain — class chants 3x: 'Variable, variable, hold the place — the unknown number fills the space!' Then Detective Dee's signature: 'Find it!'

Lesson 2: Missing Number in Addition

Goal: If $5 + ? = 12$, the missing number is 7.

Expectation: C2.2

Learning Goal: "I can find the missing number in an addition equation."

Materials: M2_balance_mat, M3_addition_puzzle_cards, M5_two_colour_counters, M8_number_line

1. Minds On (12 mins): Detective Dee Solves an Addition Case

Detective Dee found 8 cookies. After the bake-off, she has 15. How many were added? Detective Dee opens: 'Find it! Then CHECK it! That's the Detective way!'

Yesterday Detective Dee opened the VARIABLE BOXES (Day 1). Today the unknown number lives INSIDE an addition equation — ' $5 + ? = 12$ '. Detective Dee says the equal sign means SAME ON BOTH SIDES, not 'do something.' We FIND the unknown that makes both sides equal.

2. Action (25 mins): Missing-Number Addition Stations

- Whole group: solve 4 missing-addend equations together.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: solve 3-term equations like $5 + \square + 4 = 15$.
- Discuss strategies.
- Circulate to record formative evidence on the addition tracker.
- STRATEGY COMPARE: Pairs first share their OWN strategy (count-up vs count-back vs known-fact vs balance).

3. Consolidation (13 mins): Detective Dee's Addition Brag

Discuss:

- What does the variable stand for in $5 + ?$
- How can a number line help you find the missing number?
- What does it mean to make two sides EQUIVALENT?
- Can you check your answer?

Exit: Detective Dee's signature — class chants: 'Find it!'

Lesson 3: Missing Number in Subtraction

Goal: If $12 - ? = 7$, the missing number is 5.

Expectation: C2.2

Learning Goal: "I can find the missing number in a subtraction equation."

Materials: M2_balance_mat, M4_subtraction_puzzle_cards, M5_two_colour_counters, M8_number_line

1. Minds On (12 mins): Detective Dee Solves a Subtraction Case

Detective Dee had 15 cookies. She gave some away. Now she has 8. How many were given away? Detective Dee opens: 'Find it! Then CHECK it!'

Bring children to the carpet. Detective Dee + Mystery Mo puppets. 'Friends — Detective Dee has another case. Yesterday we worked on addition. Today subtraction.'

2. Action (25 mins): Missing-Number Subtraction Stations

- Whole group: solve 4 missing-number subtraction equations together.
- Send pairs to three stations (~7 min each).
- STRATEGY INVENT-AND-COMPARE: Pairs solve ONE missing-minuend card with NO assigned strategy.
- Stretch challenge at Station C: solve mixed addition AND subtraction equations like $\square - 4 = 6 + 3$.
- Discuss the addition-subtraction relationship.
- Circulate to record formative evidence on the subtraction tracker.

3. Consolidation (13 mins): Detective Dee's Subtraction Brag

Discuss:

- What's the difference between solving $15 - ?$
- When is the addition trick faster than counting back?
- What does INVERSE mean?
- Show how to check a subtraction answer.

Exit: Detective Dee's refrain — class chants: 'Find it!'

Lesson 4: Many Ways to Make 50 (and 100)

Goal: Many addition and subtraction expressions equal the same number.
 $50 = 25+25 = 40+10 = 60-10 = 75-25$.

Expectation: C2.3

Learning Goal: "I can find many equivalent expressions for the same whole number up to 100."

Materials: M2_balance_mat, M5_two_colour_counters, M6_target_cards, M8_number_line

1. Minds On (12 mins): Detective Dee's Many Ways Challenge

Detective Dee says $25 + 25$ makes 50. Mystery Mo says $40 + 10$ makes 50. Who's right?

Bring children to the carpet. Detective Dee + Mystery Mo puppets. 'Friends — Detective Dee and Mystery Mo are arguing. Dee says $25 + 25$ makes 50. Mo says $40 + 10$ makes 50. Who's right?'

2. Action (25 mins): Equivalent-Hunt Stations

- Whole group: find 6+ expressions for 75.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: include 3-term expressions ($25+15+10 = 50$).
- Discuss patterns.
- Circulate to record formative evidence on the equivalent-expressions tracker.

3. Consolidation (13 mins): Detective Dee's Equivalence Brag

Discuss:

- What does EQUIVALENT mean?
- How many ways did you find to make 50?
- Can subtraction expressions equal 50?
- Can a 3-term expression equal 50?

Exit: Each child writes ONE equivalent expression for a target of their choice on a sticky note: ' $___ = ___ + ___$ ' or ' $___ = ___ - ___$ '.

Lesson 5: Mystery Case Capstone

Goal: We can write a mystery case that uses a variable, a missing number, and an equivalent expression.

Expectation: C2.1, C2.2, C2.3

Learning Goal: "I can write a mystery case using a variable and solve it."

Materials: M2_balance_mat, M3_addition_puzzle_cards, M4_subtraction_puzzle_cards, M7_mystery_case_template

1. Minds On (12 mins): Detective Dee's Final Showcase

Now you become the detective. Pick a real-life mystery. Write the equation with a variable. Solve it.

Bring children to the carpet. Detective Dee + Mystery Mo together. 'Friends — today you become DETECTIVES. Pick a real-life mystery. Write it as an equation with a variable. Solve it.'

2. Action (25 mins): Mystery Case Capstone

- Hand out capstone templates.
- Children solve their own case.
- Stretch challenge: write a SECOND case using the same topic but a different variable position.
- Gallery walk: children swap templates and solve each other's cases.
- Whole-class share: 5-6 children share their case aloud.

3. Consolidation (13 mins): Detective Dee's Final Brag and Certificate Ceremony

Discuss:

- What's the most surprising case you saw today?
- What does the variable stand for in your case?
- How did you check your answer?
- What's something in your life you'll model after this unit?

Exit: Each child receives a Junior Detective certificate.

Date: _____

Learning Goal: I can recognise when a symbol stands for an unknown number.

1. Circle the variable in each equation.
2. Match each equation to the unknown number it stands for.
3. Write 3 of your own equations using a variable.

[illegible]

Name: _____ Date: _____

Name: _____ Date: _____

Detective Dee's Addition Cases

Learning Goal: I can find the missing number in an addition equation.

Instructions for the Student:

1. Find the missing number in each addition equation.
2. Find the missing number.
3. Write your own missing-addend case.

Part 3: Make Your Own Case

Write your own missing-addend case. Then solve it.

[illegible]

Name: _____ Date: _____

Name: _____ Date: _____

Detective Dee's Subtraction.

Learning Goal: I can find the missing number in a subtraction equation.

Instructions for the Student:

1. Find the missing number.
2. Find the missing number.
3. Write your own subtraction missing-number case.

Part 2: Subtraction to 100 (Missing-Minuend)

Find the missing number. The variable is the **STARTING** number
— use the addition trick.

[illegible]

Name: _____

Date: _____

Detective Dee's Many Ways

Learning Goal: I can find many equivalent expressions for the same whole number up to 100.

Instructions for the Student:

1. List 8 different expressions that equal 50.
2. List 6 different expressions that equal 100.
3. Pick a target number between 60 and 100.

Part 3: Pick Your Target

Pick a target number between 60 and 100. List 5 different expressions for it.

Date: _____

My Mystery Case

Learning Goal: I can write a mystery case using a variable and solve it.

Instructions for the Student:

1. Write a real-life mystery case in 1-2 sentences.
2. Translate your case into an equation.
3. Solve the equation.

Part 4: Equivalent Expression

Write 1 equivalent expression for your answer. (For example:
if your answer is 50, write $50 = 25 + 25 = 60 - 10$.)

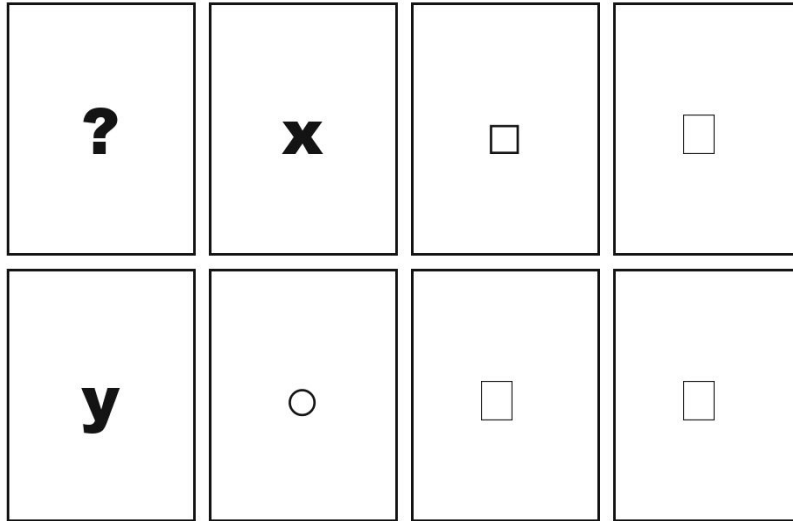
[illegible]

Magnifying Glass Variable

(M1_magnifying_cards)

24 cards each showing an equation with a missing variable.

Variable Cards



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

TEACHER PREP

1. Print 4 pages on cardstock.

2. Laminate.

3. Cut into 24 cards.

4. Sort by difficulty.

5. Distribute Lesson 1.

USE IN CLASS

Quantity: 24 cards

Lessons: 1

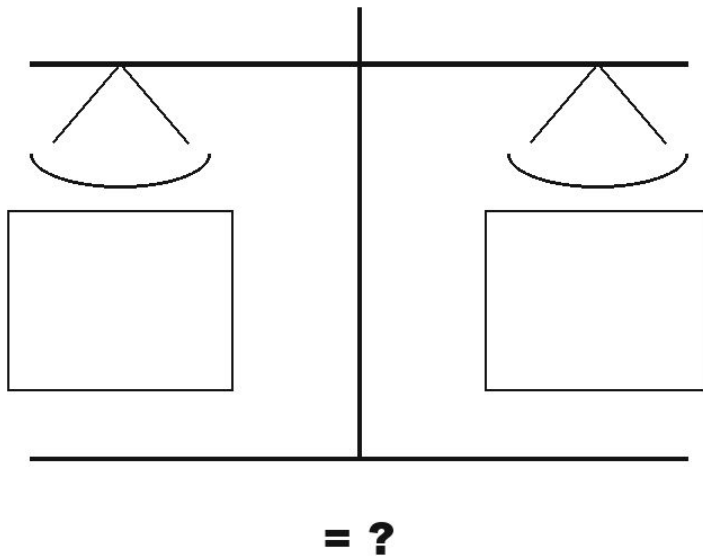
Prep: ~20 min

Balance Scale Drawing Mat

(M2_balance_mat)

Letter-size mat with a drawn balance scale (two pans).

Balance Mat



PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS

Quantity: 1 per pair
Lessons: 2, 3, 4, 5
Prep: ~12 min

TEACHER PREP

1. Print 1 mat per pair.
2. Laminate.
3. Hand out Lesson 2.
4. Use dry-erase.
5. Reuse Lessons 3-5.

Addition Missing-Number Cards

(M3_addition_puzzle_cards)

24 cards each with an addition equation and one number missing.

Addition Cards

$? + 3 = 8$	$5 + ? = 9$
$? + 4 = 12$	$10 = 6 + ?$
$? + 7 = 15$	$11 = ? + 4$
$? + 6 = 14$	$9 + ? = 13$

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 24 cards

Lessons: 2, 5

Prep: ~20 min

TEACHER PREP

1. Print 4 pages on cardstock.
2. Laminate.
3. Cut into 24 cards.
4. Sort by difficulty.
5. Distribute Lessons 2 and 5.

Subtraction Missing-Number

(M4_subtraction_puzzle_cards)

24 cards each with a subtraction equation and one number missing.

Subtraction Cards

$10 - ? = 4$	$? - 5 = 6$
$12 - ? = 7$	$? - 3 = 8$
$9 - ? = 2$	$15 - ? = 9$
$? - 6 = 5$	$8 - ? = 4$

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 24 cards

Lessons: 3, 5

Prep: ~20 min

TEACHER PREP

1. Print 4 pages on cardstock.
2. Laminate.
3. Cut into 24 cards.
4. Sort by difficulty.
5. Distribute Lessons 3 and 5.

Two-Colour Counters

(M5_two_colour_counters)

30 counters per pair for balance modelling.

Equation Table

Counters	Total	Note

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 30 per pair

Lessons: 2, 3, 4, 5

Prep: ~18 min

TEACHER PREP

1. Print 2 sheets per pair on cardstock.

2. Laminate.

3. Cut into 30 counters.

4. Sort into pair bags.

5. Distribute Lessons 2-5.

Equivalent-Target Cards

(M6_target_cards)

12 'target number' cards for the equivalent-expression hunt.

Target Cards

10	20	50	100
5	25	75	200

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

TEACHER PREP

1. Print 2 pages on cardstock.

2. Laminate.

3. Cut into 12 cards.

4. Sort by difficulty.

5. Distribute Lesson 4.

USE IN CLASS

Quantity: 12 cards

Lessons: 4

Prep: ~12 min

Mystery Case Capstone Template

(M7_mystery_case_template)

Letter-size template for Lesson 5 capstone.

1. The CASE:
2. The EQUATION:
3. SOLVE the variable:
4. Equivalent expressions:

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

USE IN CLASS

Quantity: 1 per child

Lessons: 5

Prep: ~12 min

TEACHER PREP

1. Print 1 per child.

2. Hand out Lesson 5.

3. Children fill in.

4. Collect for assessment.

5. Display gallery.

Number Line to 100

(M8_number_line)

Letter-size number line marked 0 to 100.

Number Line to 100	
1	Major Ticks At Every 5 _____
2	Minor At Every 1 _____
3	Number _____
4	Number Line to 100 _____

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 per pair

Lessons: 2, 3, 4

Prep: ~10 min

TEACHER PREP

1. Print 1 line per pair.

2. Laminate.

3. Hand out Lesson 2.

4. Use dry-erase.

5. Reuse Lessons 3-4.

Variable Anchor Chart

(M9_variable_chart)

Large classroom anchor chart showing variable symbols.

Equation Table

Variable	Stands for	Value

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall Day 1.

4. Reference all week.

5. Roll for re-use.

USE IN CLASS

Quantity: 1 chart

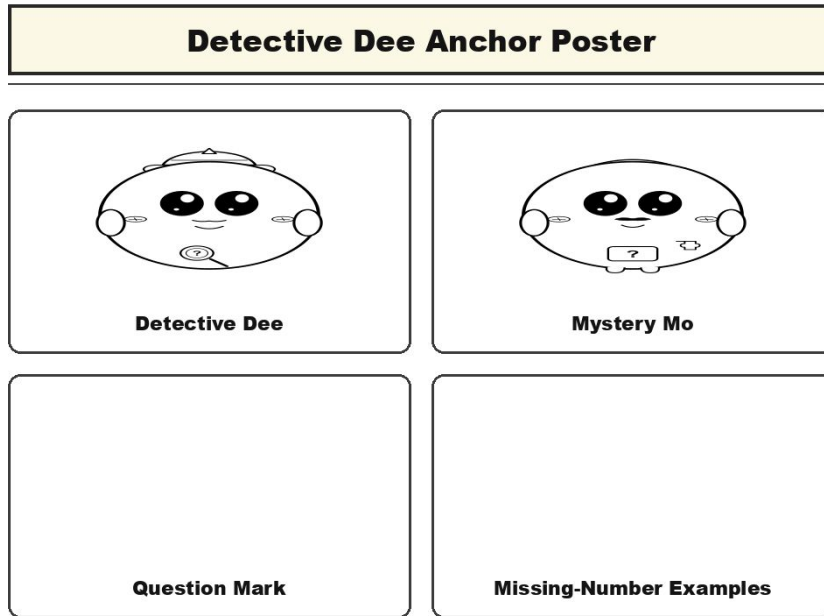
Lessons: 1, 2, 3

Prep: ~15 min

Detective Dee Anchor Poster

(M10_detective_poster)

Large classroom anchor showing Detective Dee and Mystery Mo with a missing-number case file.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 1 poster

Lessons: 1, 2, 3, 4, 5

Prep: ~18 min

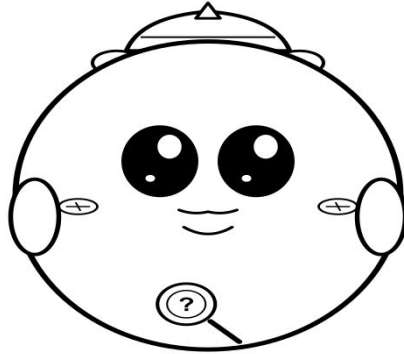
TEACHER PREP

1. Print 4 pages and tile.
2. Laminate.
3. Tape to wall Day 1.
4. Reference all week.
5. Roll for re-use.

Detective Dee Character Puppet

(char_dee_puppet)

Printable Detective Dee illustration on a popsicle stick.



Detective Dee

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

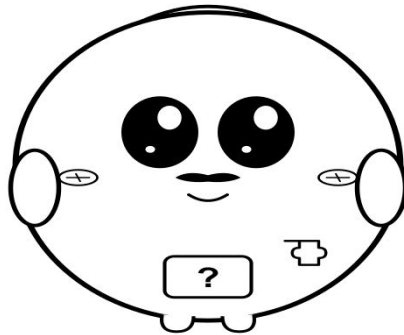
4. Optional: laminate.

5. Store flat.

Mystery Mo Character Puppet

(char_mo_puppet)

Printable Mystery Mo illustration on a popsicle stick.



Mystery Mo

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
C2.1 identify when symbols are being used as variables, and describe how they are being used	Cannot recognise a variable symbol.	Recognises variable with prompts; treats it as the answer.	Identifies variables and explains they stand for unknowns.	Uses multiple variable symbols; explains they all behave identically.
C2.2 determine what needs to be added to or subtracted from addition and subtraction expressions to make them equivalent	Cannot find missing numbers.	Finds missing addends in simple equations with prompts.	Solves addition and subtraction missing-number equations to 100.	Distinguishes missing-minuend from missing-subtrahend; checks every answer.
C2.3 identify and use equivalent relationships for whole numbers up to 100, in various contexts	Cannot generate equivalent expressions.	Generates 1-2 equivalent expressions with prompts.	Generates 5+ equivalent expressions for whole numbers to 100.	Uses both addition and subtraction; includes 3-term expressions.

Junior Detective Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day What's Missing?
unit and demonstrating the skills of a Junior Detective.
Detective Dee and Mystery Mo congratulate the student.

By completing this unit, this student has demonstrated:

- Recognising symbols as variables
- Finding missing numbers in addition equations
- Finding missing numbers in subtraction equations
- Generating equivalent expressions for numbers to 100
- Inventing and solving a mystery case

Marketplace Listing

A 5-day Ontario Grade-2 unit on Algebra (C2.1-C2.3) framed around 'What's Missing?' — Detective Dee the Math Sleuth and Mystery Mo.

Price: \$9.99 CAD

Pages: ~32

Classroom time: 250 min

Teacher prep: ~90 min

Top tags: grade-2, math, ontario-curriculum, algebra, variables, missing-numbers, equivalent-expressions, canadian

What's included:

- 5 fully-scripted lesson plans with 50-minute durations and 250 minutes of total teaching time
- 5 student worksheets aligned to each lesson
- 1 mid-unit formative assessment (Variable Check-In)
- 10 printable teacher manipulatives plus 2 character puppets: magnifying glass variable cards (24), balance scale drawing mat, addition missing-number cards...
- 1 end-of-unit reflection sheet with 5 self-assessment prompts
- 4 class observation trackers: 1 diagnostic + 3 formative
- 1 three-row summative rubric covering C2.1-C2.3 across 4 levels
- 1 summative-administration guide tying the rubric to Lesson 5 capstone
- Junior Detective certificate (one per child)
- Detailed expanded walkthroughs covering common misconceptions
- Differentiation tracks built into every lesson